From Correlation to Causation: Instrumental Variables in Practice

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GitHub Repo: https://github.com/irisyyl/Casual_Inference_causal_direction_iv

Goal

identify causal direction using instrumental variables (IV) when there is unmeasured confounding

RESEARCH QUESTION

Two-Stage least squares (2SLS) regression analysis

1. Predict X using Z
$$\widehat{X} = \alpha_0 + \alpha_1 Z + \varepsilon_1$$
2. Regress Y on predicted X
$$Y = \beta_0 + \beta_1 \widehat{X} + \varepsilon_2$$

The coefficient β1 is the causal effect of X on Y.

3 criteria

1. Relevance
2. Indepence
3. Exclusion Restriction

3 variables

Z = Instrumental Variable (IV)

X = Variable 1 (could be treatment)

Y = Variable 2 (could be outcome)

U = Unmeasured confounder (hidden)

Sensitivity & Robustness

How much hidden confounding is needed to overturn these estimates?

Approach

01.

Ordinary Least Squares (OLS)

02.

Two-Stage least squares (2SLS) regression analysis

03.

Comparison between OLS vs 2SLS

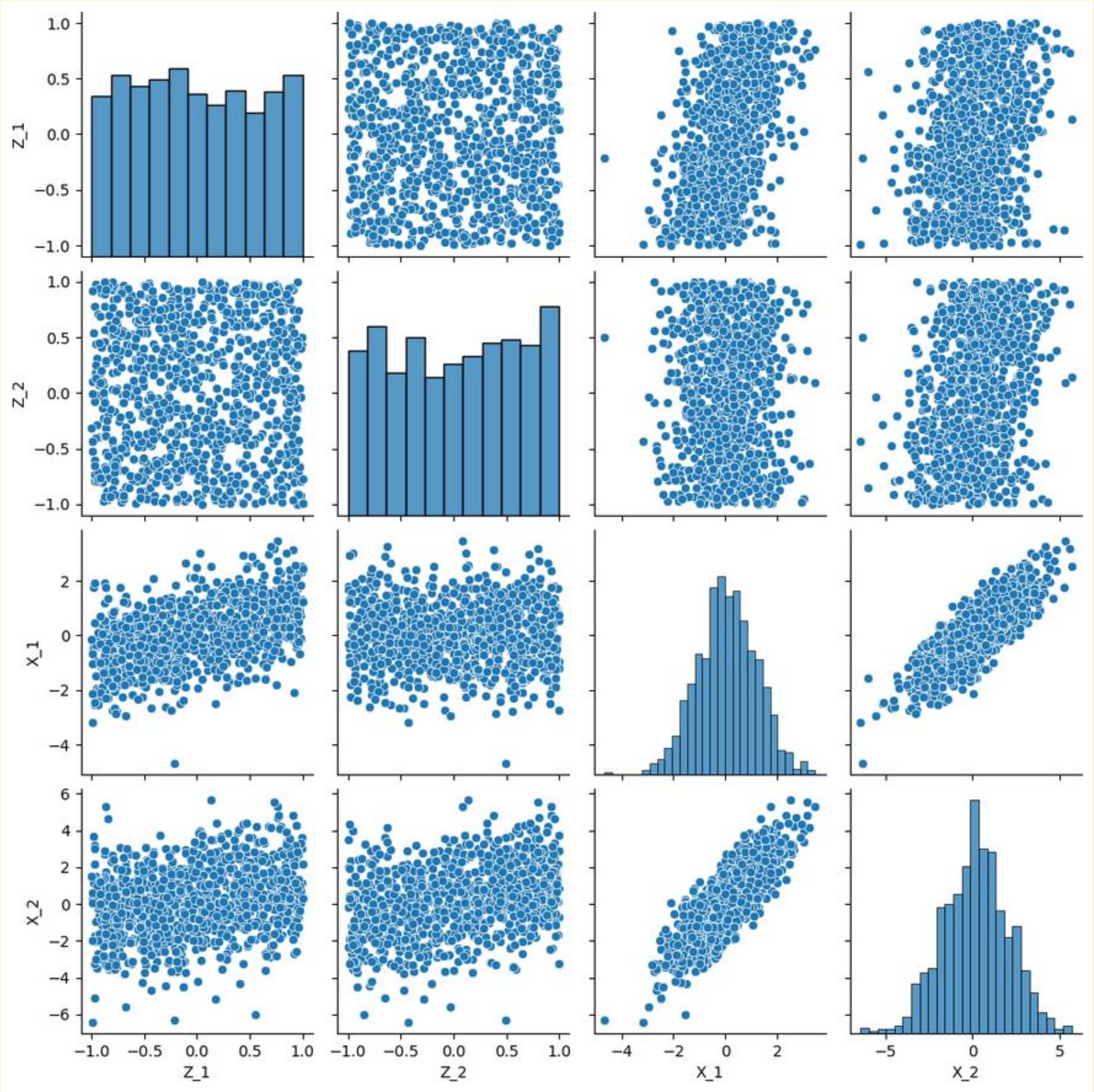
04.

Robustness Test

Scenarios

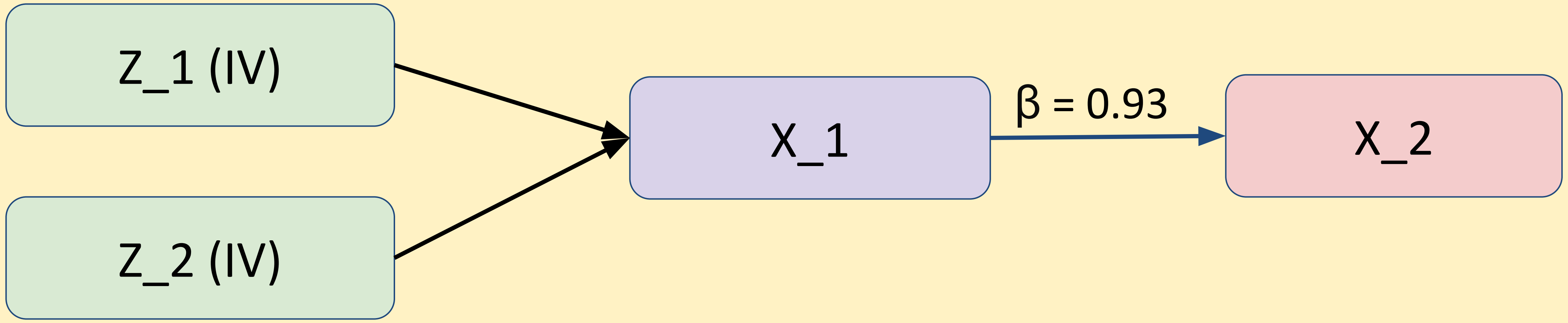

Benchmark
Clinical Trial
eCommerce
Crop yields
Marketing

Scenario 1 - synthetic IV benchmark

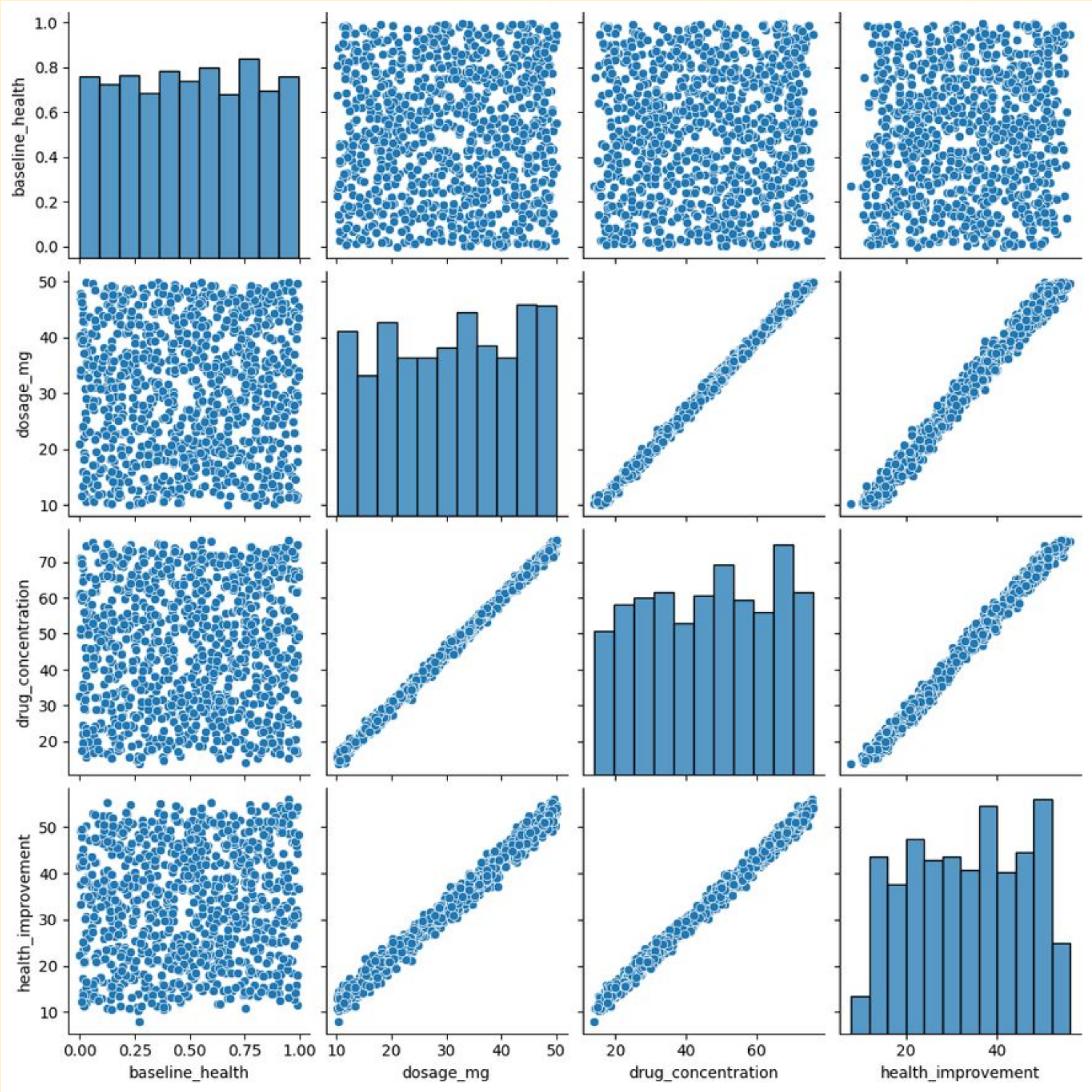


Causal Graph

IV valid

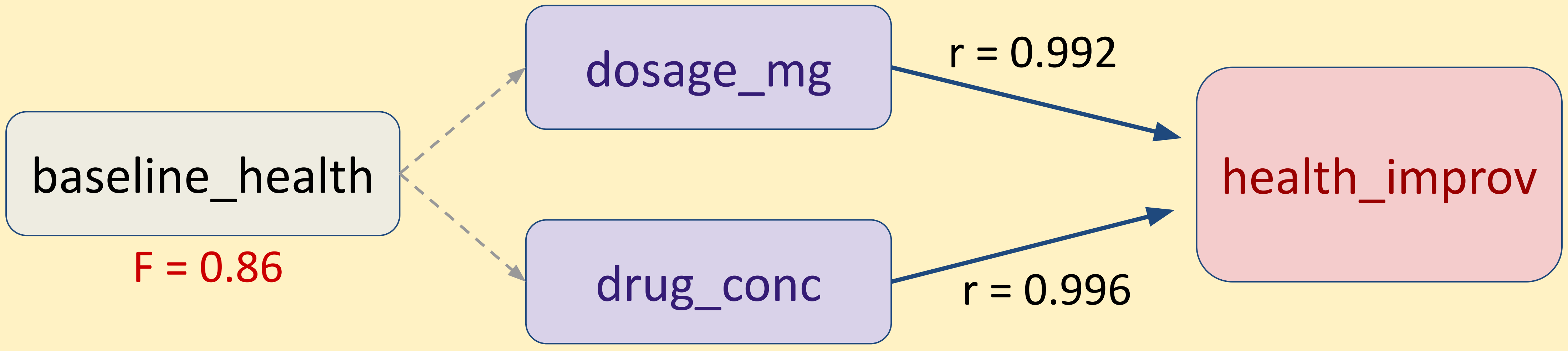


Scenario 2 - dosage & health improvement

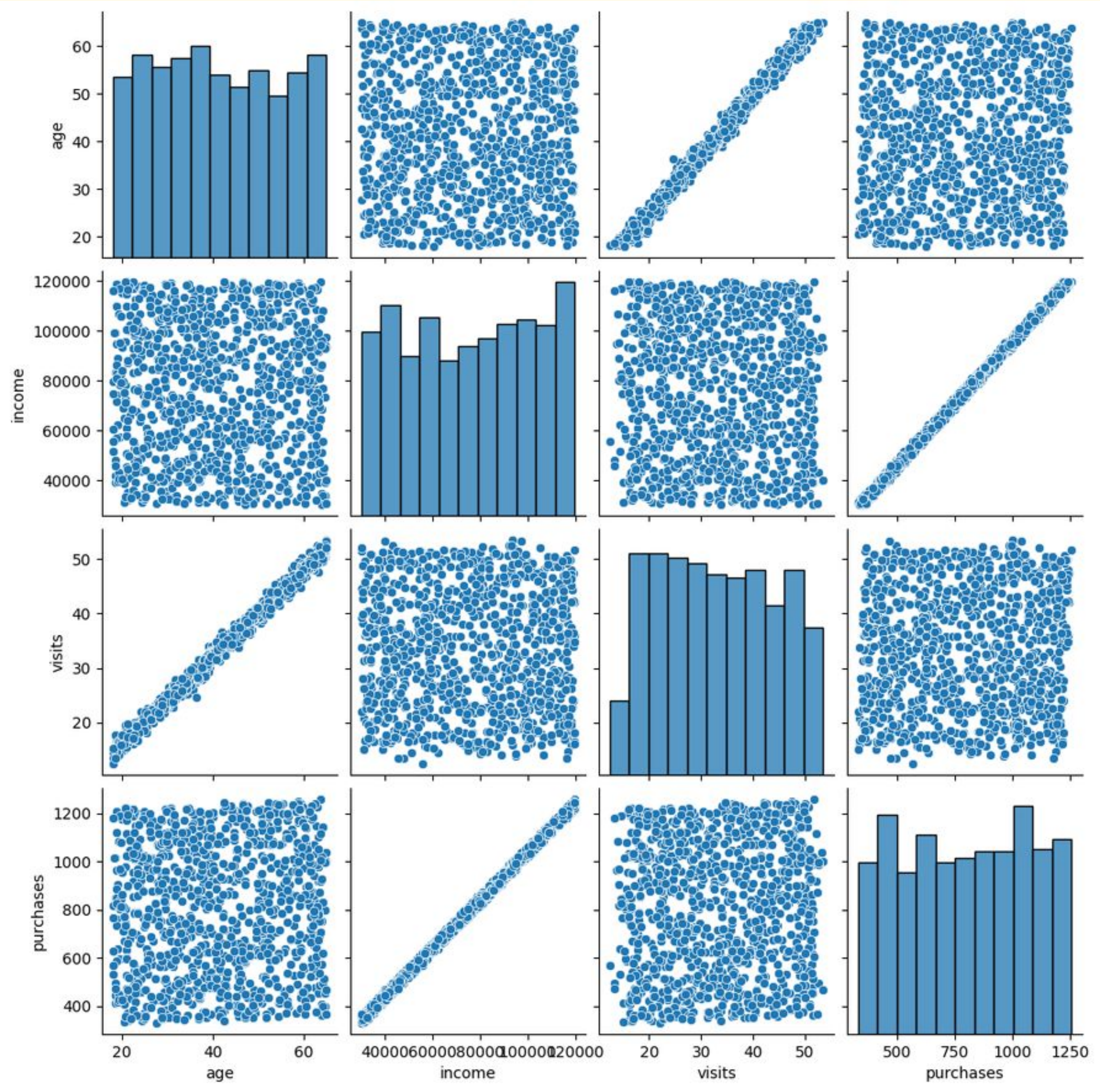


Causal Graph

IV failed



Scenario 3 - income vs purchases



Causal Graph

IV valid

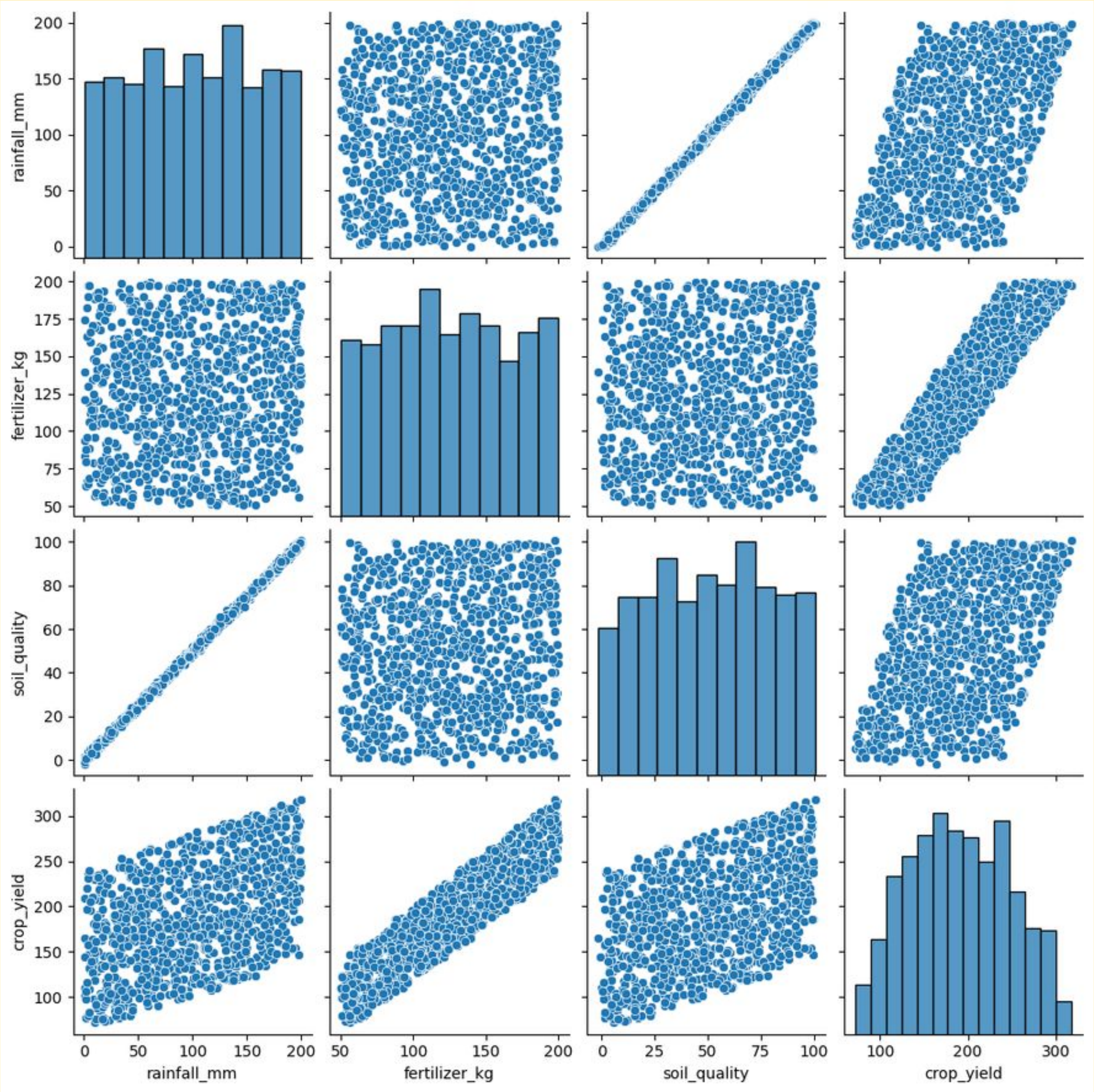
Chain 1

age —(r = 0.996)→ visit --(no effect)--> purchases

Chain 2

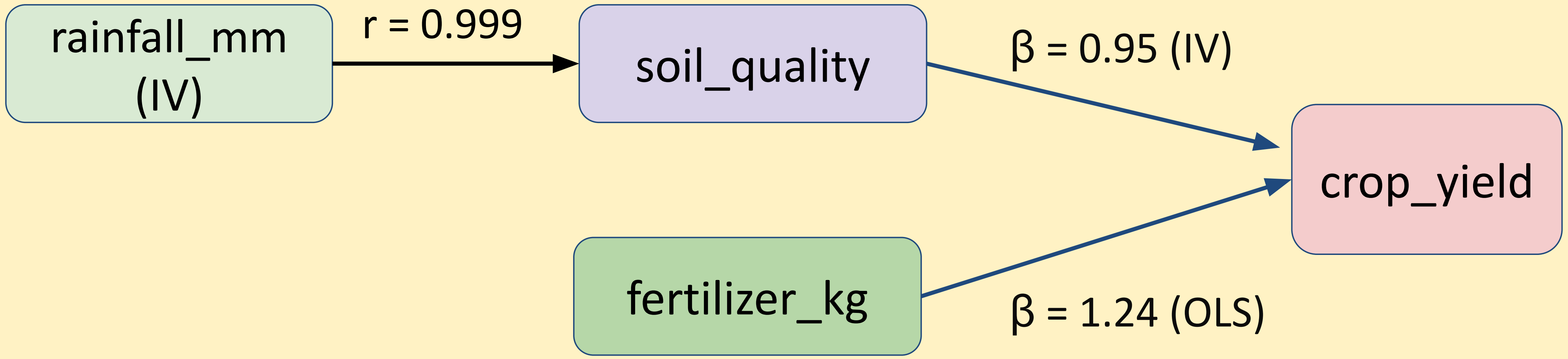
income —(β = 0.01/\$ · R^2=0.998)→ purchases

Scenario 4 - crop yield vs fertilizer

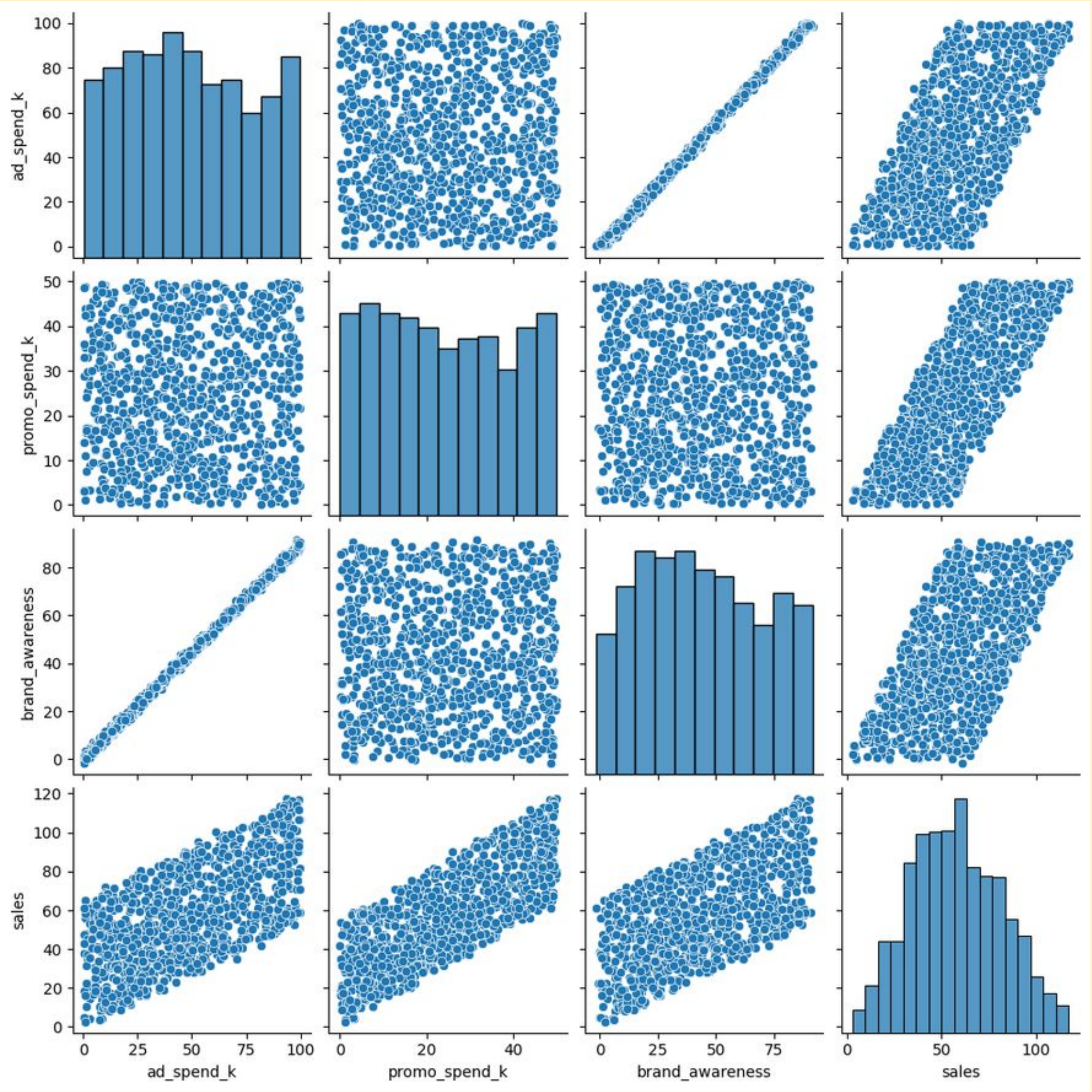


Causal Graph

IV valid

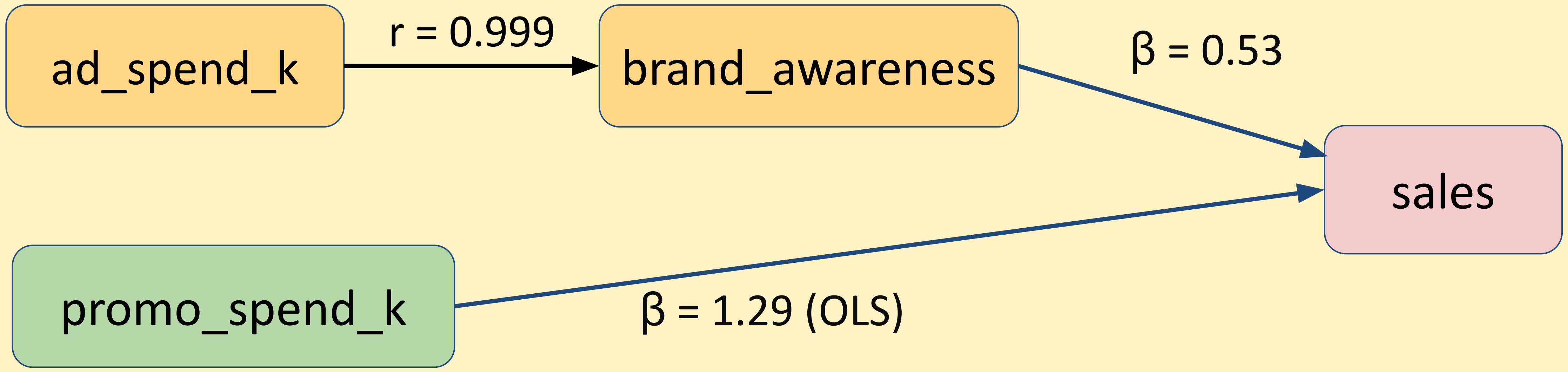


Scenario 5 - sales vs promo

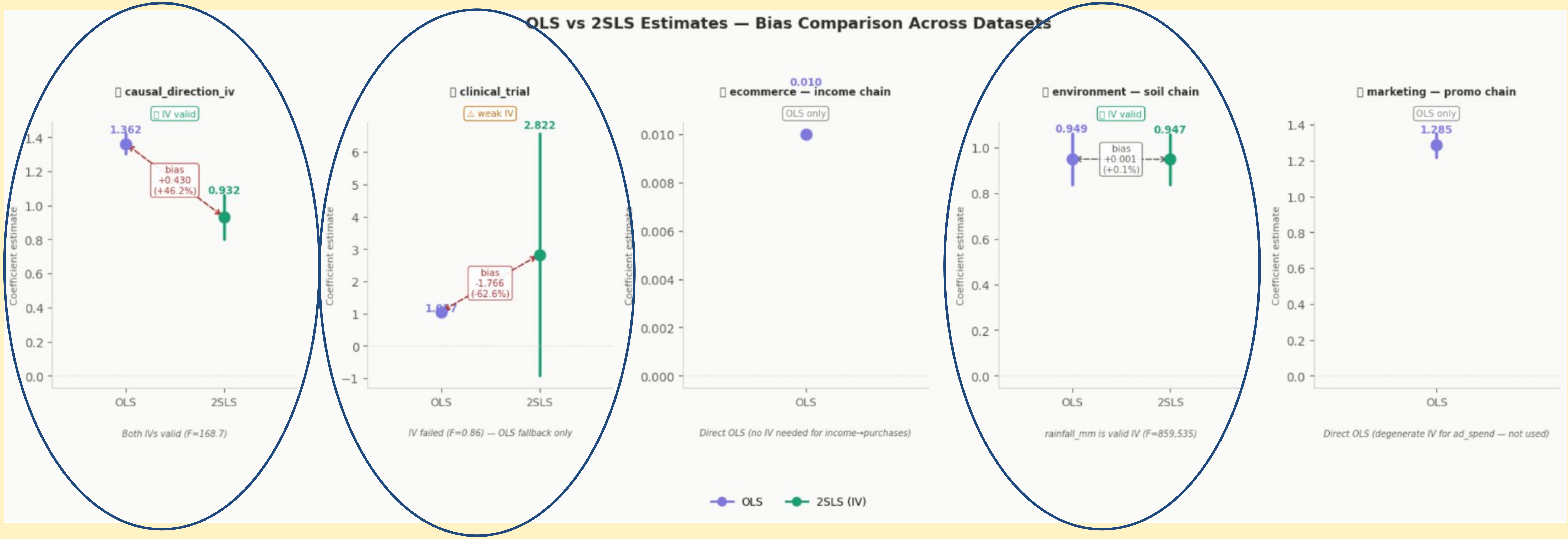


Causal Graph

Degenerate IV

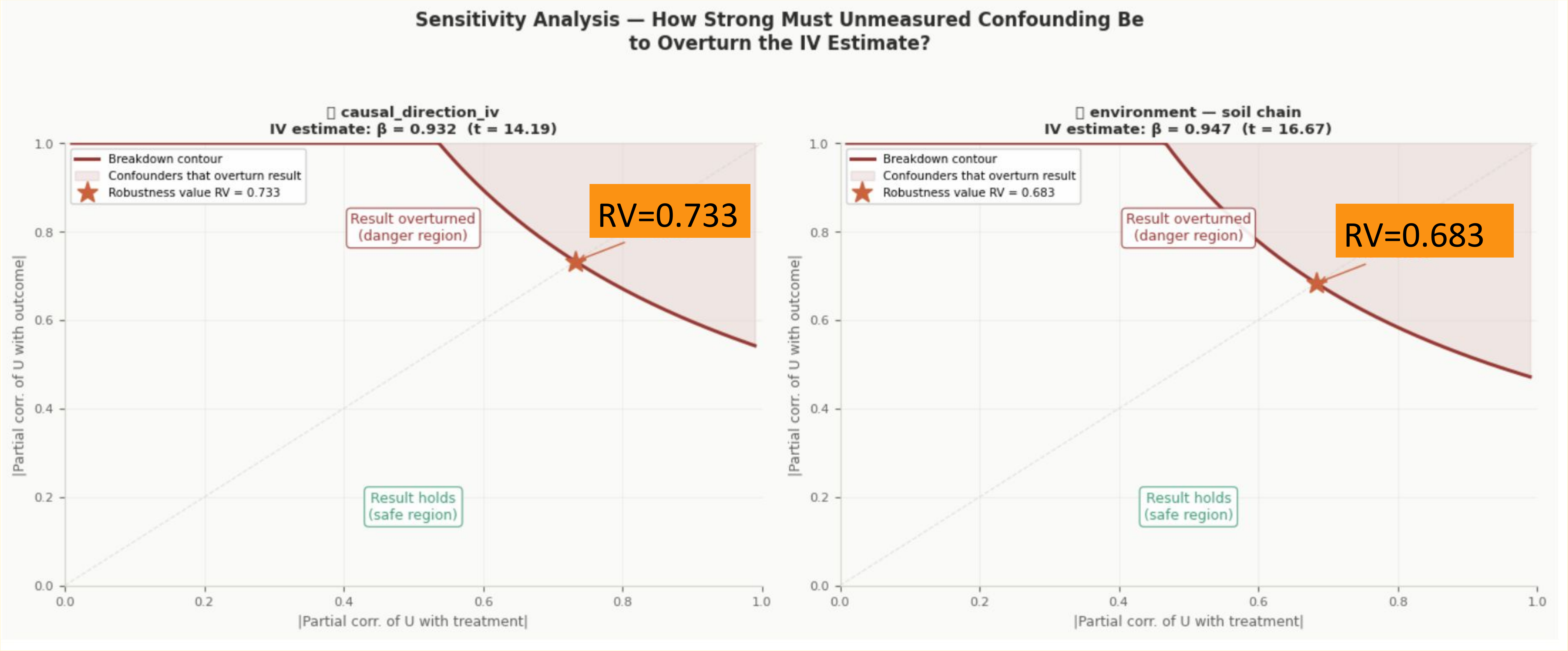


Comparison - OLS vs 2SLS

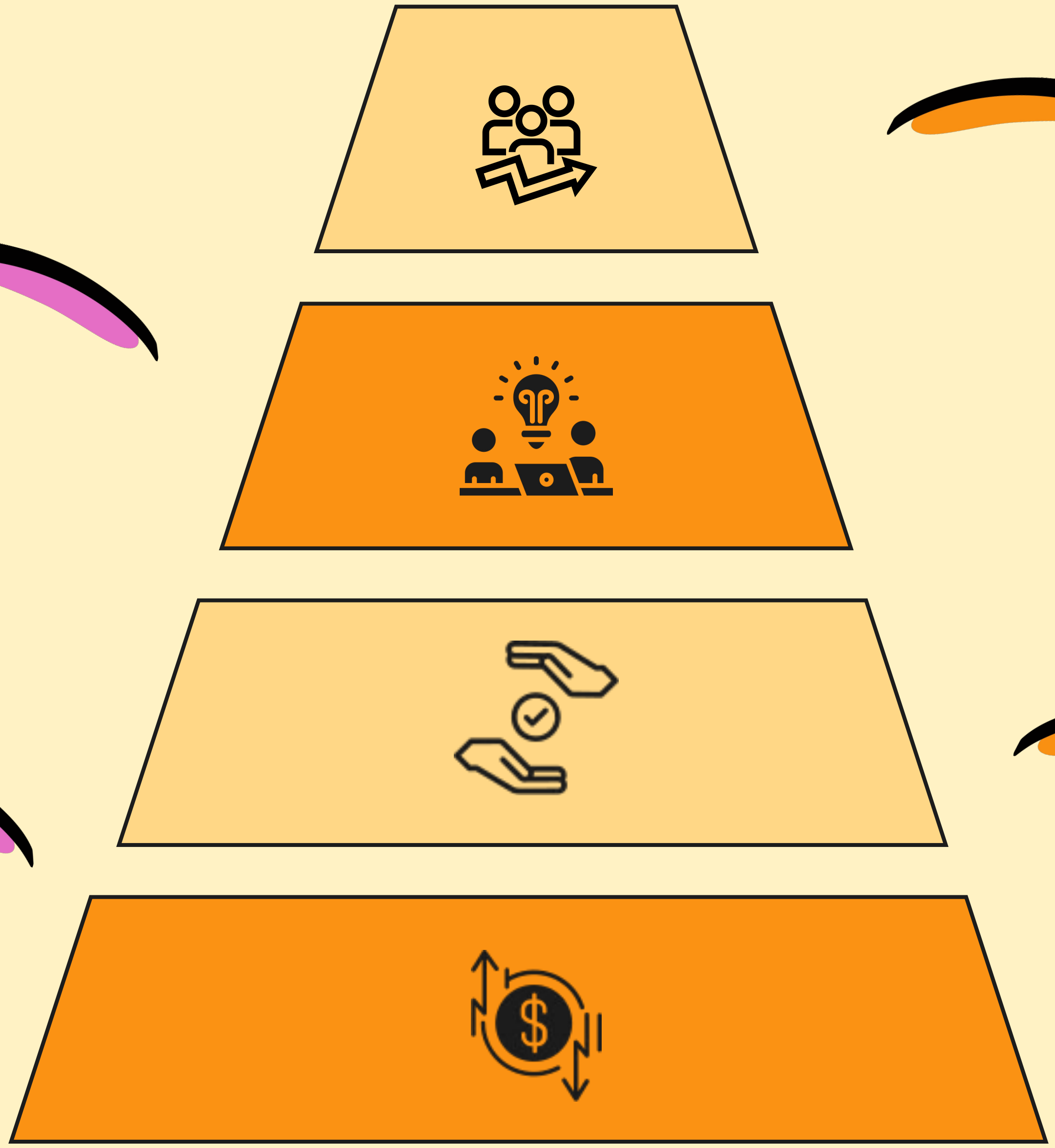


Endogeneity bias

Robustness


Sensitivity Analysis — How Strong Must Unmeasured Confounding Be to Overturn the IV Estimate?
causal_direction_iv
IV estimate: β = 0.932 (t = 14.19)
Breakdown contour
Confounders that overturn result
Robustness value RV = 0.733
Result overturned (danger region)
Result holds (safe region)
RV=0.733
|Partial corr. of U with outcome|
|Partial corr. of U with treatment|
environment — soil chain
IV estimate: β = 0.947 (t = 16.67)
Breakdown contour
Confounders that overturn result
Robustness value RV = 0.683
Result overturned (danger region)
Result holds (safe region)
RV=0.683
|Partial corr. of U with outcome|
|Partial corr. of U with treatment|

Key takeaways



1. Instrument validity is binary.

2. High correlation ≠ causal order.

3. Degenerate IVs are detectable.

4. Null results are results.

Non-linear / Limited dependent variable IV

When outcome/treatment is binary/count/nonlinear

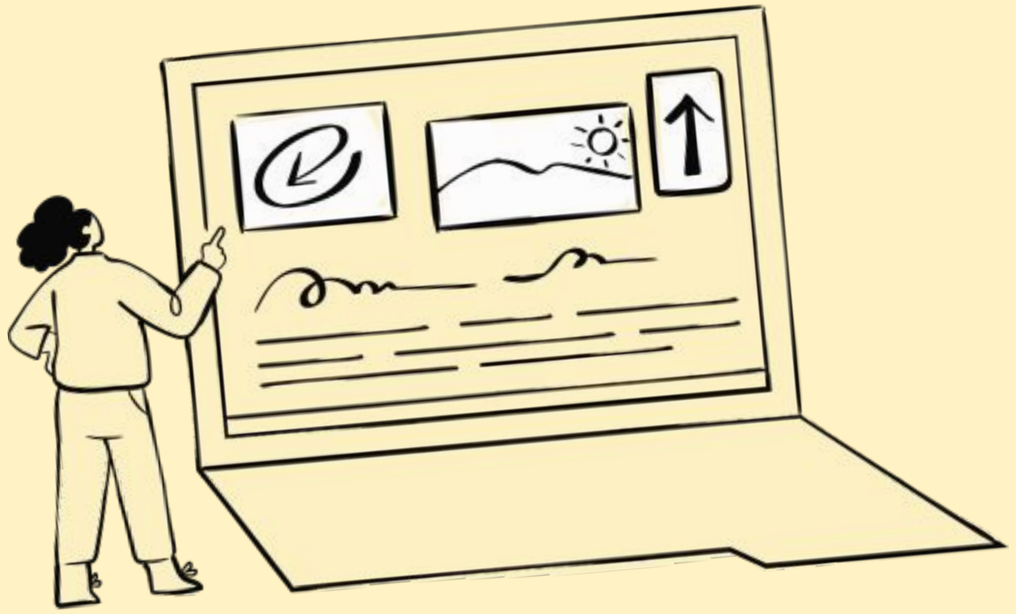
GMM (Generalized Method of Moments) IV

When there are a lot of instruments, or dealing with heteroskedasticity/cluster standard error

ML-based / DML-IV (double ML)

When there is high-dimensional data, combining with lasso/random forest

Advanced Techniques



Thank you for your attention !

Questions?